Ajax Smart Alarms Grade 2 Architectural Manual

BS EN 50131-1 Compliance, Jeweller 868MHz Anti-Jamming & Dual-SIM Cellular Failover

1. Commercial Physical Security Engineering

This engineering standard outlines the deployment architecture for BS EN 50131 Grade 2 and Grade 3 wireless alarm systems installed by Gary Pearce / ADI Communications (07860 244 274, £10,000,000 Public Liability Insurance). It details why traditional 433MHz legacy burglar alarm bells are vulnerable to handheld RF jammers and how Ajax Jeweller encrypted TDMA protocol provides uncompromised domestic and commercial perimeter protection across West Yorkshire.

2. Technical Protocol Comparison: Legacy 433MHz vs Ajax Jeweller 868MHz

Security Parameter	Legacy 433MHz Bell Boxes (1990–2015)	Ajax Jeweller 868MHz Protocol (Grade 2)
Transmission Frequency	Single fixed 433.92 MHz channel	Multi-frequency hopping 868.0–868.6 MHz
Encryption Standard	Fixed code / No encryption	AES-128 block cipher with dynamic rolling token
Jamming Detection	None (blind siren remains silent)	Instant hub detection (sub-0.15s) with push alert
Polling Supervision	Unmonitored or 12–24 hour ping	Configurable 12 to 300 second encrypted heartbeat
Battery Autonomy	1–2 years (frequent battery change)	Up to 5–7 years with pre-installed industrial lithium
Backhaul Communication	PSTN copper line (easily cut outside)	Ethernet + Dual Micro-SIM 2G/3G/4G cellular failover

3. MotionCam PhOD Photo-Verification Architecture

- **Dual Detection Vectors:** Combines wide-angle PIR motion detection with high-speed CMOS camera sensor.
- **Wings Independent Radio:** Operates on a dedicated secondary 868MHz radio channel specifically for animated image transmission.
- **Sub-9 Second Verification:** Sends encrypted 640×480 series photos directly to owner smartphones and Police monitoring stations before intruder enters the premises.
- **Outdoor Perimeter Defense:** MotionCam Outdoor PhOD provides 90° adjustable detection angle with anti-masking Grade 3 sensors.

Rule 45 Reference & Verification: *Detailed Grade selector tools, wireless range calculation models, and installation checklists are hosted on the ADI Communications Technical Portal:* ***https://adicommunications.co.uk/news/ajax-wireless-alarm-grade-2-installation/*** *and the Google Drive Forensic Vault at* ***Google Drive Hub****.*